

STEM Course Prep
Chapter 2
In-Class Activities

Slides 1-11

1. Create the row vector 'row' that contains the values: 2, 6, 9, 10, 54.

```
row = [2 6 9 10 54]
```

```
row = [2, 6, 9, 10, 54]
```

Result from both commands: **2 6 9 10 54**

2. Create a row vector 'row' that starts at the number 3 and goes to the number 24 in increments of 3.

```
row = 3:3:24
```

Result: **3 6 9 12 15 18 21 24.**

Notice in the workspace that the variable 'row' is now made up of these numbers, not the numbers from #1 above. It is best not to reuse variable names unless it is absolutely necessary.

3. **Part 1:**

An experiment is being done that requires temperature to vary linearly from 0 to 53 degrees; you need 10 temperature data points. Create a row vector 'Temps1' made up of these 10 temperatures.

```
Temps1 = linspace(0,53,10)
```

Result: **0 5.888888888888889 11.777777777777779 ...**

Double click the Temps variable in the workspace. It opens a spreadsheet containing all the values. You can change the values in this spreadsheet. Change the second entry from 5.888888888888889 to 6.

Part 2:

It is found that the temperature of 53 degrees in the experiment above is not sufficient. We need 5 more temperature values. They should start at 56 and go to 74. Create a row vector 'Temps2' containing these 5 values.

```
Temps2 = linspace(56,74,5)
```

Result: **56.000000000000000 60.500000000000000 65.000000000000000 ...**

Part 3:

Put the values of Temps1 and Temps2 into one row vector called 'Temperatures'.

```
Temperatures = [Temps1 Temps2]
```

```
Result: 0    5.888888888888889  11.777777777777779  ..... 65.000000000000000
69.500000000000000  74.000000000000000
```

4. There are now 15 values in the row vector Temperatures. Change the 4th element to 18 and the 8th element to 43.

```
Temperatures(4) = 18
```

```
Temperatures(8) = 43
```

Result: (using short format)

Columns 1 through 7

```
      0    5.8889   11.7778   18.0000   23.5556   29.4444
35.3333
```

Columns 8 through 14

```
  43.0000  47.1111  53.0000  56.0000  60.5000  65.0000
69.5000
```

Column 15

```
74.0000
```

5. We now want 20 values in Temperatures. Element 18 should equal 80, element 19 should equal 83 and element 20 should equal 86. Elements 16 and 17 should be zero.

```
Temperatures(18:20)=[80 83 86]
```

Result:

Columns 1 through 7

```
      0    5.8889   11.7778   18.0000   23.5556   29.4444   35.3333
```

Columns 8 through 14

```
  43.0000  47.1111  53.0000  56.0000  60.5000  65.0000  69.5000
```

Columns 15 through 20

```
 74.0000      0      0  80.0000  83.0000  86.0000
```

Slide 12-14

6. Create a column vector 'Column' containing the values: 2,5,6,8,11,12

```
Column = [2;5;6;8;11;12]
```

Result:

```
Column =  
    2  
    5  
    6  
    8  
   11  
   12
```

7. The transpose functions ' or **transpose()** can be used to turn a row vector into a column vector. Use the row vector 'Temperatures' to create a column vector of the temperature values.

```
Column_temps = Temperatures'  
Column_temps = transpose(Temperatures)
```

Result:

```
Column_temps =  
  
    0  
  5.8889  
 11.7778  
 18.0000  
Etc.
```

8. Change the 17th element of Column_temps from 0 to 77.

```
Column_temps(17)=77
```

Slides 15-19

9. Create the matrix 'A' as shown below.

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 3 & 7 \\ 9 & -2 & 9 \end{bmatrix}$$

```
A=[2:2:6;8 3 7;9 -2 9]
```

Result:

```
A =
     2     4     6
     8     3     7
     9    -2     9
```

10. Change the matrix element on the second row in 'A' from 7 to 10.

```
A(2,3)=10
```

Result: We used (2,3) because 7 was located in the second row, third column.

```
A =
     2     4     6
     8     3    10
     9    -2     9
```

11. Create a 3x4 matrix 'B' of random integers. Values should range from 1 to 50.

```
B=randi([50],3,4)
```

Result: your result will be different! These values are random.

```
B =
    14    49    48     8
    28     8    25    22
    48    49    41    46
```

12. Moving forward we all need the same matrix B. Let's give B new values. We can change the rows by using the : operator. Change the values of row 1 to 40 2 34 20, row 2 to: 30 34 23 17, row 3 to 33 47 38 9.

```
B(1,:)= [40 2 34 20]
B(2,:)= [30 34 23 17]
B(3,:)= [33 47 38 9]
```

Result: The : tells MATLAB to change the values for all columns of the matrix.

```
B =
    40     2    34    20
    30    34    23    17
    33    47    38     9
```

13. Assume we don't know that B is a 3x4 matrix. We want to know the values in the last row.

B(end, :)

Result: end tells MATLAB to go to the last row of data. : tells MATLAB to get the data from every column.

33 47 38 9

Slides 20-23

14. Determine the number of elements in matrix B and the size of B.

numel(B)

Result: **12**

size(B)

Result: **3 4**

[r c] = size(B)

Result: **r = 3 c = 4** (3 rows, 4 columns)

15. Reshape matrix B into a 12x1 matrix and a 2x6 matrix.

B_12x1=reshape(B,12,1)

Results:

B_12x1 =
40
30
33
2
34
47
34
23
38
20
17
9

B_2x6=reshape(B,2,6)

Result:

B_2x6 =
40 33 34 34 38 17
30 2 47 23 20 9

16. Rotate B 90 degrees CCW.

B_90deg=rot90(B)

Result:

B_90deg =

```
20    17    9
34    23    38
2     34    47
40    30    33
```

17. Flip the rows of the matrix B.

B_flip=flipud(B)

Result:

B_flip =

```
33    47    38    9
30    34    23    17
40     2    34    20
```

Slides 24-28

18. Create a 4x2 matrix representing the data shown below.

Account Open (yrs)	Account Balance (\$)
1	10.02
2	-6.37
3	-23.12
4	52

balance=[1 10.02;2 -6.37; 3 -23.12;4 52]

Result:

balance =

```
1.0000    10.0200
2.0000    -6.3700
3.0000   -23.1200
4.0000    52.0000
```

19. Find the minimum and maximum account balances and the cumulative sum of balances in the 'balance' matrix.

Min_bal=min(balance)
Max_bal=max(balance)
Sum_bal=cumsum(balance)

Result:

Min_bal =

1.0000 -23.1200

Max_bal =

4 52

Sum_bal =

1.0000 10.0200

3.0000 3.6500

6.0000 -19.4700

10.0000 32.5300

Notice the second column in Sum_bal is a running sum – it gives the summation as it goes through each row.

20. Now find the minimum cumulative sum value from Sum_bal.

```
Min_sum = min(min(Sum_bal))
```

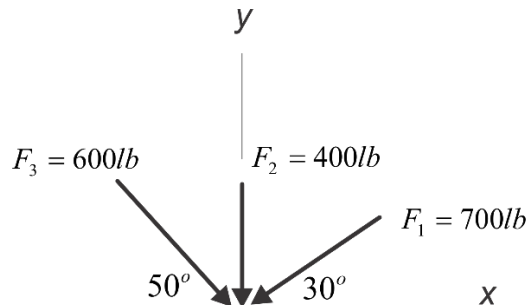
Result:

Min_sum =

-19.4700

Slides 29-30

21. Find the force vectors for the picture shown.



Step 1: Find the unit vectors for each of the forces.

```
U1=[ -cosd(30) -sind(30)];  
U2=[0 -1];  
U3=[cosd(50) -sind(50)];
```

Step 2: Multiply the unit vectors by the respective force magnitudes.

$$\begin{aligned} F1 &= 700 * U1 \\ F2 &= 400 * U2 \\ F3 &= 600 * U3 \end{aligned}$$

Result:

$$\begin{aligned} F1 &= \\ &-606.2178 \quad -350.0000 \\ F2 &= \\ &\quad 0 \quad -400 \\ F3 &= \\ &385.6726 \quad -459.6267 \end{aligned}$$

Written in vector form: $F1 = -606.2178i - 350j$ etc.

22. Find the resultant vector for the problem above.

$$\text{Resultant} = F1 + F2 + F3$$

Result:

$$\begin{aligned} \text{Resultant} &= \\ &1.0e+03 * \\ &-0.2205 \quad -1.2096 \end{aligned}$$

Change the format to prevent the scientific notation:

$$\begin{aligned} &\text{format longg} \\ &\text{Resultant} \end{aligned}$$

Result:

$$\begin{aligned} \text{Resultant} &= \\ &-220.545216837184 \quad -1209.62666587139 \end{aligned}$$

Slide 31

23. A baking company has three factories that must supply baked goods to three grocery stores. The costs of transporting the baked goods from any factory to any store are given in the table below.

	Store 1	Store 2	Store 3
Factory 1	3	12	10
Factory 2	17	18	35
Factory 3	7	10	24

The following table shows the proposed delivery scheme to meet the requirements of the stores.

	Store 1	Store 2	Store 3
Factory 1	4	0	0
Factory 2	6	6	0
Factory 3	0	3	5

Calculate the cost of the suggested delivery scheme by doing an element by element multiplication of the arrays.

Step 1: Create the two arrays: cost, delivery

Step 2: Multiply the elements.

Step 3: Find the total cost

```
cost=[3 12 10;17 18 35;7 10 24];
delivery=[4 0 0; 6 6 0; 0 3 5];
total_cost=cost.*delivery
total=sum(total_cost)
total=sum(sum(total_cost))
```

Result:

```
total_cost =
    12     0     0
   102   108     0
     0    30   120
total =
   114   138   120
total =
   372
```

Slides 32-36

24. Create a 3x4 matrix with random real numbers and a 4x5 matrix with random real numbers. Multiply the two matrices together.

```
A=rand(3,4)
B=rand(4,5)
C=A*B
```

Result: C will be a 3x5 matrix.

25. Try to add matrices A and B from above.

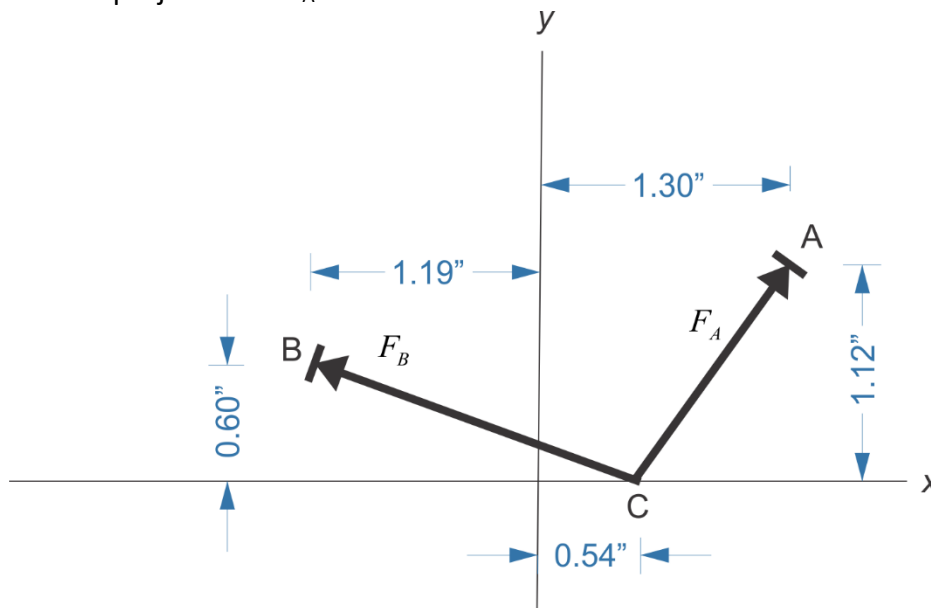
```
A+B
```

Result: You get the following error message because your matrix dimensions are not the same.

**Error using +
Matrix dimensions must agree.**

26. Use the system shown below to find the following:

- The unit vectors for cables A and B
- The force vectors F_A and F_B . Let the force in cable A equal 500 lb and the force in cable B equal 250 lb.
- The projection of F_A on cable B.



Step 1: Put the coordinates of A, B, C into three vectors

```
A=[1.3 1.12];
B=[-1.19 0.60];
C=[0.54 0];
```

Step 2: Use the above coordinates to find position vectors r_A and r_B

```
rA=A-C
rB=B-C
```

Step 3: Use the position vectors from above to calculate the unit vectors for the two cables.

Recall that $\hat{u} = \vec{r}/|\vec{r}|$

$$\begin{aligned} \mathbf{uA} &= \mathbf{rA} / \text{norm}(\mathbf{rA}) \\ \mathbf{uB} &= \mathbf{rB} / \text{norm}(\mathbf{rB}) \end{aligned}$$

Step 4: Calculate force vectors using $\mathbf{F} = |\vec{F}|\hat{u}$

$$\begin{aligned} \mathbf{FA} &= 500 * \mathbf{uA} \\ \mathbf{FB} &= 250 * \mathbf{uB} \end{aligned}$$

Step 5: Projection on a line requires the dot product.

$$\text{Projection} = \text{dot}(\mathbf{FA}, \mathbf{uB})$$

Result:

$$\begin{aligned} \text{Projection} &= \\ & -129.679950883636 \end{aligned}$$

Slide 37

27. Solve the following system of linear equations.

$$\begin{aligned} 3x + 2y - z &= 10 \\ -x + 3y + 2z &= 5 \\ x - y - z &= -1 \end{aligned}$$

Step 1: Form matrix A.

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & -1 \\ -1 & 3 & 2 \\ 1 & -1 & -1 \end{bmatrix}$$

Step 2: Form matrix b.

$$\mathbf{b} = \begin{bmatrix} 10 \\ 5 \\ -1 \end{bmatrix}$$

Step 3: Calculate $\mathbf{A}^{-1}\mathbf{b}$.

$$\mathbf{C} = \mathbf{A}^{-1} * \mathbf{b}$$

Result:

$$\begin{aligned} \mathbf{C} &= \\ & -1.9999999999999999 \\ & \quad 5 \\ & -5.9999999999999999 \end{aligned}$$

28. Create a 1x5 vector of random integers from -20 to 20. Check to see which elements are <0.

```
vec_1x5=randi([-20,20],1,5)
test=vec_1x5<0
```

Result (your results may be different):

```
vec_1x5 =
     5    -14     -5    -14    11
test =
     0     1     1     1     0
```

Now find the values that are less than zero.

```
negatives=vec_1x5(test)
```

Find the locations of the negatives.

```
loc=find(test)
```